CSE563: Unit III Study Notes

Probability and Statistics

School of Computer Science & Engineering

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1 Unit III: Probability and Statistics

Course Outcome CO3: apply the concepts of probability and statistics for inference and regression analysis.

Probability is the mathematics of uncertainty; statistics is the discipline of drawing conclusions from data under that uncertainty. Every machine-learning model is a probabilistic statement, every A/B test is a hypothesis test, every spam filter is Bayes' theorem, and every trend line is regression.

1.1 Basic Probability

Definition 1.1. A **random experiment** has an uncertain outcome. The **sample space** S is the set of all possible outcomes; an **event** $E \subseteq S$ is any subset. Probability is a function P on events obeying Kolmogorov's axioms:

1. $0 \le P(E) \le 1$; 2. $P(S) = 1$; 3. if $E_1, E_2, \ldots$ are mutually exclusive, $P(E_1 \cup E_2 \cup \ldots) = \sum P(E_i)$.

Three Ways to Assign Probability

- **Classical:** equally likely outcomes, $P(E) = |E|/|S|$. Fair die: $P(\text{even}) = 3/6$.
- **Empirical (frequentist):** $P(E) = \lim_{n\to\infty}$ (number of times E occurs)$/n$. Click-through rate from logs.
- **Subjective (Bayesian):** degree of belief, updated by evidence. "70% chance this server fails this month."

Formula & Mathematical Relation: Consequences of the Axioms

$$P(\overline{E}) = 1 - P(E) \tag{1}$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad \text{(addition rule)} \tag{2}$$
$$P(\emptyset) = 0, \qquad A \subseteq B \Rightarrow P(A) \le P(B) \tag{3}$$

In the addition rule, drop the last term when A and B are mutually exclusive.

Formula & Mathematical Relation: Counting (needed for classical probability)

Permutations of r from n (order matters): $P(n,r) = \dfrac{n!}{(n-r)!}$. Combinations (order irrelevant): $\binom{n}{r} = \dfrac{n!}{r!(n-r)!}$. Product rule: k independent choices with $n_1, \ldots, n_k$ options give $n_1 n_2 \cdots n_k$ outcomes.

Worked Example: Classical Probability with Counting

A 4-digit PIN. $P(\text{all digits distinct}) = \dfrac{10 \cdot 9 \cdot 8 \cdot 7}{10^4} = 0.504$. Two dice: $P(\text{sum} = 7) = 6/36 = 1/6$; $P(\text{sum} \ge 10) = (3 + 2 + 1)/36 = 1/6$. Drawing 2 cards from 52 without replacement: $P(\text{both aces}) = \binom{4}{2} / \binom{52}{2} = 6/1326 = 1/221$. $P(\text{at least one six in 4 rolls}) = 1 - (5/6)^4 = 1 - 0.482 = 0.518$ (complement is almost always the fastest route to "at least one").

1.2 Compound Probability

Definition 1.2 (Conditional Probability)**.** The probability of A given that B has occurred:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0.$$

Conditioning *shrinks the sample space* to B.

Formula & Mathematical Relation: Multiplication Rule, Independence, Total Probability

$$P(A \cap B) = P(A \mid B)\,P(B) = P(B \mid A)\,P(A) \quad \text{(multiplication rule)} \tag{4}$$

$$A, B \text{ independent} \iff P(A \cap B) = P(A)P(B) \quad \text{(equivalently } P(A \mid B) = P(A)\text{)} \tag{5}$$

$$P(A) = \sum_i P(A \mid B_i)\,P(B_i) \quad \text{(law of total probability)} \tag{6}$$

In the law of total probability the events $B_1, B_2, \ldots$ form a partition of S. Chain rule for several events: $P(A_1 \cap A_2 \cap A_3) = P(A_1)\,P(A_2 \mid A_1)\,P(A_3 \mid A_1 \cap A_2)$.

Common Pitfall & Misconception: Mutually Exclusive ≠ Independent

Mutually exclusive: $P(A \cap B) = 0$ (they cannot both happen). Independent: $P(A \cap B) = P(A)P(B)$ (one tells you nothing about the other). Two events with positive probability that are mutually exclusive are *maximally dependent*: if A happened, B certainly did not.

Worked Example: Conditional Probability and Independence

In a company, 60% of employees use Linux (L), 50% use Docker (D), and 40% use both. $P(D \mid L) = 0.4/0.6 = 0.667$. $P(L \mid D) = 0.4/0.5 = 0.8$. Independent? $P(L)P(D) = 0.30 \neq 0.40 = P(L \cap D)$: no, they are positively associated. $P(L \cup D) = 0.6 + 0.5 - 0.4 = 0.7$, so 30% use neither.

Formula & Mathematical Relation: Bayes' Theorem

For a partition $B_1, \ldots, B_k$ of the sample space:

$$P(B_j \mid A) = \frac{P(A \mid B_j)\,P(B_j)}{P(A)} = \frac{P(A \mid B_j)\,P(B_j)}{\sum_i P(A \mid B_i)\,P(B_i)}. \tag{7}$$

It **reverses the direction of conditioning**: from "probability of evidence given cause" to "probability of cause given evidence".

Worked Example: The Medical Test (Base-Rate Fallacy)

A disease has prevalence 1%. A test has sensitivity $P(+ \mid D) = 0.99$ and specificity $P(- \mid \overline{D}) = 0.95$ (so false-positive rate 0.05). A random person tests positive. Probability they have the disease?

$$P(D \mid +) = \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.05 \times 0.99} = \frac{0.0099}{0.0099 + 0.0495} = \frac{0.0099}{0.0594} = 0.167.$$

Only 16.7%, despite a "99% accurate" test, because the healthy population is 99 times larger and produces most of the positives. Same arithmetic governs spam filters, intrusion detection and fraud alerts: a rare-event detector with a small false-positive rate still generates mostly

false alarms.

1.3 Random Variables and Probability Distributions

Definition 1.3. A **random variable** X is a function from the sample space to $\mathbb{R}$. It is **discrete** if it takes countably many values (described by a *probability mass function* $p(x) = P(X = x)$) and **continuous** if it takes values in an interval (described by a *probability density function* $f(x)$ with $P(a \le X \le b) = \int_a^b f(x)\,dx$). The **cumulative distribution function** is $F(x) = P(X \le x)$ in both cases.

Formula & Mathematical Relation: Expectation and Variance

$$E[X] = \mu = \sum_x x\,p(x) \quad \text{or} \quad \int x f(x)\,dx \tag{8}$$

$$\mathrm{Var}(X) = \sigma^2 = E[(X-\mu)^2] = E[X^2] - \mu^2 \tag{9}$$

$$E[aX+b] = aE[X] + b, \qquad \mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X) \tag{10}$$

$$E[X+Y] = E[X] + E[Y] \text{ always;} \quad \mathrm{Var}(X+Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y \text{ if independent.} \tag{11}$$

Standard deviation $\sigma = \sqrt{\mathrm{Var}(X)}$ has the same units as X.

Distribution	PMF / PDF	Mean	Variance	Models
Bernoulli(p)	$P(X{=}1) = p$, $P(X{=}0) = 1{-}p$	p	$p(1-p)$	One yes/no trial
Binomial(n, p)	$\binom{n}{x}p^x(1-p)^{n-x}$, $x = 0..n$	np	$np(1-p)$	# successes in n independent trials
Geometric(p)	$(1-p)^{x-1}p$, $x = 1, 2, \ldots$	$1/p$	$(1-p)/p^2$	Trials until first success
Poisson(λ)	$e^{-\lambda}\lambda^x/x!$, $x = 0, 1, \ldots$	λ	λ	Counts of rare events per interval (requests/sec, bugs/kLOC)
Uniform(a, b)	$1/(b-a)$ on $[a, b]$	$(a+b)/2$	$(b-a)^2/12$	"No idea" between two limits; `rand()`
Exponential(λ)	$\lambda e^{-\lambda x}$, $x \ge 0$	$1/\lambda$	$1/\lambda^2$	Waiting time between Poisson events; memoryless
Normal(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2	Sums/averages of many effects (CLT); measurement noise

Table 1: Standard distributions. Binomial with large n and small $p \approx$ Poisson(np); Binomial with large $n \approx$ Normal($np, np(1-p)$).

Worked Example: Discrete Distributions

Binomial. A link fails with probability 0.2; five packets are sent independently. $P(\text{exactly 2 fail}) = \binom{5}{2}(0.2)^2(0.8)^3 = 10 \times 0.04 \times 0.512 = 0.2048$. Expected failures $= 1$, variance $= 0.8$.
Poisson. A server receives on average 3 requests per second. $P(0 \text{ in a second}) = e^{-3} = 0.0498$; $P(\le 1) = e^{-3}(1+3) = 0.199$; $P(\ge 2) = 0.801$.
Geometric. $P(\text{first success on the 3rd trial})$ with $p = 0.25$: $(0.75)^2(0.25) = 0.1406$. Expected trials $= 4$.

Formula & Mathematical Relation: Standardising the Normal Distribution

If $X \sim N(\mu, \sigma^2)$ then $Z = \dfrac{X-\mu}{\sigma} \sim N(0,1)$. Look up $\Phi(z) = P(Z \le z)$ in the standard normal table. Values to memorise:

z	0.5	1.0	1.28	1.5	1.645	1.96	2.0	2.58
$\Phi(z)$	0.6915	0.8413	0.90	0.9332	0.95	0.975	0.9772	0.995

Symmetry: $\Phi(-z) = 1 - \Phi(z)$. Empirical rule: 68% within $\pm 1\sigma$, 95% within $\pm 2\sigma$, 99.7% within $\pm 3\sigma$.

Worked Example: Normal Probabilities

Response times are $N(\mu = 170\,\text{ms}, \sigma = 8)$. $P(X > 182) = P(Z > 1.5) = 1 - 0.9332 = 0.0668$. $P(158 < X < 178) = P(-1.5 < Z < 1) = 0.8413 - (1 - 0.9332) = 0.7745$. The 90th percentile: $z = 1.28 \Rightarrow x = 170 + 1.28 \times 8 = 180.2$ ms (an SLA "p90 latency").

1.4 Frequentist Inference

Population, Sample, Estimator

We want a **parameter** of a population (mean μ, proportion p) but can only observe a **sample**. A **statistic** computed from the sample (sample mean $\bar{x}$, sample proportion $\hat{p}$, sample variance $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$) serves as an **estimator**. The frequentist view: the parameter is a fixed unknown constant; randomness lives in the sample. Statements are about what would happen over *repeated sampling*.

Formula & Mathematical Relation: Central Limit Theorem and the Standard Error

For i.i.d. samples of size n from any distribution with mean μ and variance σ^2, the sample mean satisfies

$$\bar{X} \;\approx\; N\!\left(\mu,\; \frac{\sigma^2}{n}\right) \quad \text{for large } n \text{ (rule of thumb } n \ge 30). \tag{12}$$

The **standard error** $\text{SE} = \sigma/\sqrt{n}$ shrinks with $\sqrt{n}$: to halve the uncertainty, quadruple the data. For a proportion, $\text{SE} = \sqrt{p(1-p)/n}$.

Formula & Mathematical Relation: Confidence Intervals

A $100(1-\alpha)\%$ confidence interval for μ:

$$\sigma \text{ known (or } n \text{ large}): \quad \bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \qquad (z_{0.025} = 1.96,\; z_{0.05} = 1.645,\; z_{0.005} = 2.58) \tag{13}$$

$$\sigma \text{ unknown, small } n: \quad \bar{x} \pm t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}} \qquad (t \text{ table, } n-1 \text{ degrees of freedom}) \tag{14}$$

$$\text{proportion}: \quad \hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \tag{15}$$

Meaning: if we repeated the sampling many times, 95% of the intervals so constructed would contain μ. It is *not* "there is a 95% probability μ is in this interval"—that is a Bayesian statement.

Hypothesis Testing in Five Steps

1. State H_0 (null: "no effect", e.g. $\mu = \mu_0$) and H_1 (alternative: $\mu \neq \mu_0$ two-sided, or $\mu > \mu_0$ one-sided).
2. Choose significance level α (usually 0.05) = tolerated probability of a **Type I error** (rejecting a true H_0). A **Type II error** (β) is failing to reject a false H_0; **power** $= 1 - \beta$.
3. Compute the test statistic: $z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ or $t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$.
4. Compute the **p-value**: probability, *if* H_0 *were true*, of a statistic at least this extreme. Two-sided: $p = 2(1 - \Phi(|z|))$.
5. Decide: $p < \alpha \Rightarrow$ reject H_0 ("statistically significant"). Otherwise *fail to reject*—never "accept H_0".

Worked Example: z-Test and Confidence Interval

A CPU's mean job time is claimed to be $\mu_0 = 50$ ms with known $\sigma = 6$. A sample of $n = 36$ jobs has $\bar{x} = 52$.
95% CI: $52 \pm 1.96 \times 6/\sqrt{36} = 52 \pm 1.96 = [50.04,\ 53.96]$.
Test $H_0 : \mu = 50$ *vs* $H_1 : \mu \neq 50$*:* $z = (52 - 50)/(6/6) = 2.0$. $p = 2(1 - 0.9772) = 0.0456 < 0.05$. Reject H_0: the mean job time differs from 50 ms. (Consistent with the CI, which excludes 50.) If H_1 were one-sided ($\mu > 50$): $p = 0.0228$.

Worked Example: t-Test (Small Sample, σ Unknown)

$n = 16$ compile times: $\bar{x} = 10.5$ s, $s = 2$. Test $H_0 : \mu = 10$ vs $H_1 : \mu \neq 10$ at $\alpha = 0.05$. $t = (10.5 - 10)/(2/\sqrt{16}) = 0.5/0.5 = 1.0$, with 15 df. Critical value $t_{0.025,15} = 2.131$. Since $|1.0| < 2.131$, fail to reject. The data do not show a change. 95% CI: $10.5 \pm 2.131 \times 0.5 = [9.43,\ 11.57]$, which contains 10.

Exam Tip & Key takeaway: Other Common Tests (know what they are for)

- **Two-sample t-test:** compare means of two groups (A/B test of two page designs).
- **Paired t-test:** same subjects measured twice (before/after optimisation).
- **Chi-square test of independence:** two categorical variables (browser vs. purchase). Statistic $\chi^2 = \sum(O - E)^2/E$.
- **ANOVA:** compare means of three or more groups.

p-values are not the probability that H_0 is true; "not significant" is not "no effect"; and testing many hypotheses inflates false positives (Bonferroni: use α/m).

1.5 Bayesian Inference

The Bayesian Recipe

Treat the parameter θ as a random variable with a **prior** distribution $P(\theta)$ encoding what we believe before data. Observe data D. Update:

$$\underbrace{P(\theta \mid D)}_{\text{posterior}} = \frac{\overbrace{P(D \mid \theta)}^{\text{likelihood}}\ \overbrace{P(\theta)}^{\text{prior}}}{\underbrace{P(D)}_{\text{evidence}}} \propto P(D \mid \theta)\, P(\theta).$$

The posterior is a full distribution over θ; report its mean, its mode (**MAP** estimate) or a

credible interval (a 95% credible interval *does* mean "95% probability θ lies here"). Today's posterior is tomorrow's prior: Bayesian inference is naturally sequential / online.

Formula & Mathematical Relation: Beta–Binomial Conjugate Update

For a success probability θ with prior $\text{Beta}(\alpha, \beta)$ and data of k successes in n trials:

$$\text{posterior} = \text{Beta}(\alpha{+}k,\ \beta{+}n{-}k), \qquad E[\theta \mid D] = \frac{\alpha + k}{\alpha + \beta + n}, \qquad \text{MAP} = \frac{\alpha + k - 1}{\alpha + \beta + n - 2}. \tag{16}$$

$\text{Beta}(1, 1)$ is the uniform ("know nothing") prior; α, β act like *pseudo-counts* of prior successes and failures. Other conjugate pairs: Gamma–Poisson, Normal–Normal.

Worked Example: Is the Coin Biased? Frequentist vs. Bayesian

We flip a coin 10 times and see 7 heads.
Frequentist (MLE): $\hat{\theta} = 7/10 = 0.7$. 95% CI: $0.7 \pm 1.96\sqrt{0.7 \times 0.3/10} = 0.7 \pm 0.284 = [0.42, 0.98]$.
Test $H_0 : \theta = 0.5$: $z = (0.7 - 0.5)/\sqrt{0.25/10} = 1.26$, $p = 0.21$; not significant.
Bayesian: prior $\text{Beta}(2, 2)$ (mild belief the coin is fair). Posterior $= \text{Beta}(2{+}7,\ 2{+}3) = \text{Beta}(9, 5)$.
Posterior mean $= 9/14 = 0.643$; MAP $= 8/12 = 0.667$. The prior has pulled the estimate from 0.7 toward 0.5 (*shrinkage*); with 1000 flips the prior would be negligible and both approaches agree. With a uniform prior $\text{Beta}(1, 1)$ the MAP equals the MLE, 0.7.

Question	Frequentist	Bayesian
What is θ?	Fixed unknown constant	Random variable with a distribution
Where is the randomness?	In the data (repeated sampling)	In our uncertainty about θ
Point estimate	MLE: $\arg\max_\theta P(D \mid \theta)$	MAP: $\arg\max_\theta P(D \mid \theta)P(\theta)$; or posterior mean
Interval	Confidence interval (long-run coverage)	Credible interval (direct probability)
Prior knowledge	Not used	Explicit, via prior
Small data	Wide CIs, unstable	Regularised by prior
Cost	Cheap, closed-form	May need sampling (MCMC)
ML examples	Least squares, logistic regression, t-tests	Naive Bayes, Bayesian optimisation, Kalman filter, spam filters

Table 2: Frequentist versus Bayesian inference. MAP with a Gaussian prior is exactly L2-regularised (ridge) regression—the two schools meet in practice.

Where Computer Science Uses This: Naive Bayes Classifier

For class C and features $x_1, \ldots, x_p$ assumed conditionally independent given the class:

$$P(C \mid \mathbf{x}) \propto P(C) \prod_{i=1}^{p} P(x_i \mid C).$$

Spam example: $P(\text{spam}) = 0.4$; word "free" appears in 60% of spam and 5% of ham; "meeting" in 2% of spam and 30% of ham. Email contains both. Spam score $\propto 0.4 \times 0.6 \times 0.02 = 0.0048$; ham score $\propto 0.6 \times 0.05 \times 0.30 = 0.009$. Normalise: $P(\text{spam} \mid \mathbf{x}) = 0.0048/(0.0048 + 0.009) = 0.35$. Classified as ham. Add-one (Laplace) smoothing avoids zero probabilities for unseen words—a $\text{Beta}(1, 1)$-style prior in disguise.

1.6 Regression Analysis

Correlation First

The **Pearson correlation coefficient**

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \in [-1, 1]$$

measures the strength of *linear* association. $r = \pm 1$: perfect line; $r = 0$: no linear relation (there may still be a non-linear one). Correlation is not causation: ice-cream sales and drowning are correlated through temperature (a *confounder*).

Formula & Mathematical Relation: Simple Linear Regression by Least Squares

Model: $y = \beta_0 + \beta_1 x + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$. Minimising $\sum(y_i - \hat{y}_i)^2$ gives

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \tag{17}$$

The fitted line always passes through $(\bar{x}, \bar{y})$ and the residuals $e_i = y_i - \hat{y}_i$ sum to zero. Note $\hat{\beta}_1 = r \, \frac{s_y}{s_x}$.

Goodness of fit:

$$\text{SST} = \sum(y_i - \bar{y})^2, \quad \text{SSE} = \sum(y_i - \hat{y}_i)^2, \quad \text{SSR} = \text{SST} - \text{SSE}, \quad R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}. \tag{18}$$

R^2 = fraction of the variance in y explained by the line; for simple regression $R^2 = r^2$.

Worked Example: Fitting and Interpreting a Regression Line

Advertising spend x (lakh rupees) and sales y (thousand units) for five months: $x = 1, 2, 3, 4, 5$; $y = 1, 3, 3, 5, 8$.
$\bar{x} = 3$, $\bar{y} = 4$. $S_{xy} = (-2)(-3) + (-1)(-1) + 0 + (1)(1) + (2)(4) = 6 + 1 + 0 + 1 + 8 = 16$; $S_{xx} = 4 + 1 + 0 + 1 + 4 = 10$; $S_{yy} = 9 + 1 + 1 + 1 + 16 = 28$.
$\hat{\beta}_1 = 16/10 = 1.6$; $\hat{\beta}_0 = 4 - 1.6 \times 3 = -0.8$. Line: $\hat{y} = -0.8 + 1.6x$.
Fitted values: $0.8, 2.4, 4.0, 5.6, 7.2$. Residuals: $0.2, 0.6, -1.0, -0.6, 0.8$ (sum $= 0$ ✓). SSE $= 0.04 + 0.36 + 1 + 0.36 + 0.64 = 2.4$. SST $= 28$. $R^2 = 1 - 2.4/28 = 0.914$; $r = 16/\sqrt{10 \times 28} = 0.956$ and $r^2 = 0.914$ ✓.
Interpretation: each additional lakh of advertising is associated with 1.6 thousand more units; 91% of the variation in sales is explained by spend. Predicting at $x = 6$: $\hat{y} = 8.8$ (extrapolation—use with caution). The intercept -0.8 is meaningless physically (negative sales at zero spend); it is just where the line crosses.

Formula & Mathematical Relation: Multiple Linear Regression (Matrix Form)

With p predictors, $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ where X is $n \times (p+1)$ (a column of ones for the intercept). The least-squares solution (the *normal equations*) is

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}. \tag{19}$$

This is Unit II linear algebra applied to Unit III statistics. $X^T X$ is singular when predictors are perfectly collinear. Use **adjusted** $R^2 = 1 - (1 - R^2)\frac{n-1}{n-p-1}$ to compare models with different numbers of predictors.

Inference on the Slope

Is x really related to y? Test $H_0 : \beta_1 = 0$ with $t = \hat{\beta}_1 / \operatorname{SE}(\hat{\beta}_1)$, where $\operatorname{SE}(\hat{\beta}_1) = \sqrt{\dfrac{\operatorname{SSE}/(n-2)}{S_{xx}}}$, on $n-2$ df. In the example: $\operatorname{SE} = \sqrt{(2.4/3)/10} = \sqrt{0.08} = 0.283$, $t = 1.6/0.283 = 5.66 > t_{0.025,3} = 3.182$, so the slope is significantly non-zero.

Common Pitfall & Misconception: Regression Assumptions and Abuses

- **Linearity, independence of errors, constant variance, normal errors (LINE).** Check with a residual plot: a curve means non-linearity; a funnel means heteroscedasticity.
- **Outliers** with extreme x have high leverage and can swing the line.
- **Extrapolation** beyond the observed x range is unsupported by the data.
- R^2 **always rises** when you add predictors—use adjusted R^2 or a validation set.
- Regression shows *association*, not causation, unless the data come from a randomised experiment.

Where Computer Science Uses This: Logistic Regression (Regression for Classification)

When $y \in \{0, 1\}$, model the probability with the sigmoid from Unit I: $P(y = 1 \mid \mathbf{x}) = \sigma(\boldsymbol{\beta}^T \mathbf{x}) = \dfrac{1}{1 + e^{-\boldsymbol{\beta}^T \mathbf{x}}}$. Equivalently, the log-odds are linear: $\ln \dfrac{p}{1-p} = \boldsymbol{\beta}^T \mathbf{x}$. Coefficients are fitted by maximum likelihood (no closed form; gradient descent). A coefficient β_j means: one unit of x_j multiplies the odds by e^{β_j}.

1.7 Unit III Quick-Revision Sheet

Exam Tip & Key takeaway: Ten Things to Carry Into the Exam

1. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; "at least one" $\Rightarrow$ use the complement.
2. $P(A \mid B) = P(A \cap B)/P(B)$; independent $\iff$ $P(A \cap B) = P(A)P(B)$; mutually exclusive $\neq$ independent.
3. Bayes: $P(B_j \mid A) = P(A \mid B_j)P(B_j) / \sum_i P(A \mid B_i)P(B_i)$. Always compute the denominator with total probability.
4. Binomial mean np, var $np(1-p)$; Poisson mean = var = λ; Normal: standardise $z = (x-\mu)/\sigma$.
5. $\Phi(1.96) = 0.975$, $\Phi(1.645) = 0.95$, $\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$.
6. CLT: $\bar{X} \approx N(\mu, \sigma^2/n)$; SE $= \sigma/\sqrt{n}$. CI $= \bar{x} \pm z_{\alpha/2}\sigma/\sqrt{n}$ (or t with s, $n-1$ df).
7. Hypothesis test: H_0, α, statistic, p-value, decide. Type I = false alarm (α); Type II = miss (β).
8. Posterior $\propto$ likelihood $\times$ prior; Beta(α, β) + (k of n) $\to$ Beta$(\alpha + k, \beta + n - k)$; MAP vs MLE.
9. $\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, $R^2 = 1 - \operatorname{SSE}/\operatorname{SST} = r^2$.
10. Multiple regression: $\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}$. Correlation $\neq$ causation.